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# Experimental Emulation of LEO Downlink OFLs Affected by Turbulence and Pointing Jitter

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***Abstract*—Optical space networks are envisioned as key elements of next-generation networks, leveraging the vast bandwidth of the optical domain and the wide coverage of space networks to achieve global broadband connectivity. However, the integration of terrestrial and satellite networks via optical feeder links (OFLs) is challenged by atmospheric turbulence and pointing jitter, which cause random power fading. To assess the suitability of commercial fiber-optic hardware for OFLs, this paper introduces an experimental testbed for the emulation of turbulence- and pointing jitter-induced fading. The developed testbed is used to assess the performance of commercial 10 Gigabit Ethernet transceivers under turbulence-induced scintillation and various pointing jitter levels, emulating a downlink transmission from a low Earth orbit satellite.**

***Index Terms*—Free Space Optics, Optical Feeder Link, Fading Testbed, Scintillation, Pointing Jitter**

## I. Introduction

The transition from the 5th generation (5G) of mobile networks to the 6th generation (6G) will rely on groundbreaking technologies able to support wider bandwidths, faster data rates, and higher frequencies required to sustain emerging applications. To address the lack of global coverage that characterizes 5G, the terrestrial network will be extended to a seamlessly integrated three-dimensional architecture. The integration of airborne and spaceborne nodes with terrestrial ones will expand connectivity to remote areas and blind spots, while simultaneously enhancing network flexibility and resilience at reduced costs [1].

Satellite constellations are key elements for providing ubiquitous Internet access, taking a step forward in bridging the digital divide. However, current radio frequency (RF) -based satellite communications face bandwidth limitations, interference, and security vulnerabilities. To address these issues, the concept of fiber-through-the-air via free space optics (FSO) offers a compelling alternative [2]. FSO represents a green and cost-effective solution for high-data-rate long- and ultra-long-distance data transmission. Thus, it is particularly suitable for establishing high-speed optical feeder links (OFLs) connecting ground stations to satellites. Compared to RF systems, space laser communications offer superior speed, capacity, and security, as well as smaller and lighter terminals [3].

While FSO offers significant advantages, it is susceptible to atmospheric impairments such as absorption, scattering, hydrometeors, and turbulence. Absorption and scattering attenuate the propagating laser beam. Nonetheless, the atmospheric channel features low-absorption transmission windows both in the visible and near infrared bands, enabling effective laser beam transmission at specific wavelengths. Previous demonstrations of space-based laser communications mainly took place in the 850, 1064, and 1550 nm windows [4]. The suitability of the C band (1530-1565 nm), combined with the already mature optical fiber-based technology available at this frequency range, facilitates the OFLs development [3], [5].

On the one hand, hydrometeors such as fog and clouds can cause substantial attenuation, potentially making the link temporarily unavailable. On the other hand, turbulence induces time-varying link losses, i.e., scintillation, with fluctuation intensity varying with the elevation angle and coherence times in the millisecond range. The worst-case scintillation occurs at lower elevations, while the coherence time depends on the transverse wind speed and the relative motion between the satellite and the ground station [6], [7]. Given these considerations, low Earth orbit (LEO) satellites experience variable fading conditions. Nonetheless, they offer lower latencies, path losses, as well as lower production and launch costs, making them attractive for satellite communication systems [4]. Successful OFL demonstrations and planned LEO missions highlight a growing interest in compact, power-efficient laser communication terminals for mega-constellations of small satellites such as CubeSats and nanosats [4].

Pointing error, or jitter, is another source of fading, causing instability in the laser beam direction relative to the line of sight. This instability, stemming from mechanical vibrations, turbulence-induced beam wander, and tracking system noise, significantly reduces the received power due to the narrow divergence and Gaussian profile of laser beams [6].

Unlike fiber optic links, which face dispersion and non-linear effects, OFLs are susceptible to fading, potentially causing significant frame losses at high data rates. As fiber-

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based equipment is not designed for fading mitigation, its performance must be evaluated under these conditions. The outcomes of these analyses will enable the development and implementation of suitable countermeasures to ensure the desired link availability [5].

Laboratory emulation enables controlled and repeatable OFL performance verification, bypassing unpredictable atmospheric conditions. Building on this concept, we previously developed a fading testbed that emulates turbulence-induced scintillation by replicating received power fluctuations through modulated beam intensity in a single-mode fiber (SMF) [7]. By employing this testbed, we conducted a preliminary 10 Gigabit Ethernet (10GbE) LEO downlink assessment at fixed elevation angles [7], [8]. This paper expands upon our prior research by emulating both variable turbulence- and pointing jitter-induced fading, which typically occur during a LEO satellite pass. Specifically, we experimentally investigate the impact of pointing jitter on the frame error rate using our developed testbed and 10GbE fiber-based commercial transceivers.

While fading testbeds exist in the literature that emulate OFL affected by both turbulence and pointing jitter [9], [10], our approach offers distinct advantages. Roth et al. [9] use disturbance mirrors to reproduce local platform jitter, while an electro-optic modulator applies scintillation time series generated by means of complex wave optics simulations. In contrast, our testbed relies on a computationally efficient method for generating both turbulence- and pointing jitter-related time series, which yields long fading vectors at very low computational cost. Although similar algorithms are used by Downey et al. [10], their focus differs: they evaluate the performance of custom photon counting receivers, while we analyze commercial equipment developed for terrestrial networks. Furthermore, their experiments are limited to specific fading conditions. In contrast, we replicate the dynamic scintillation experienced during a LEO satellite pass and examine the impact of varying pointing jitter levels on link performance.

The paper is organized as follows. Section II introduces the developed testbed together with the employed fading generation algorithms. Section III defines the emulated OFL scenario and presents the generated fading time series. Section IV outlines and discusses the obtained experimental results. Finally, conclusions are drawn in Section V.

## II. Experimental testbed for OFL emulation

To assess the performance of an emulated 10GbE OFL affected by scintillation- and jitter-induced fading, we set up the experimental testbed illustrated in Fig. 1. Using the iPerf2 tool in client mode, the client PC generates User Datagram Protocol (UDP) traffic at a target rate of 1 Gbit/s. The UDP datagrams are sent to the server PC, where iPerf2 is executed in server mode. For each transmission test, the server returns the percentage of lost datagrams that, due to the one-to-one encapsulation of UDP datagrams within Ethernet frames, directly corresponds to the frame loss rate. Socket buffer sizes have been optimized to prevent unintended datagram drops.

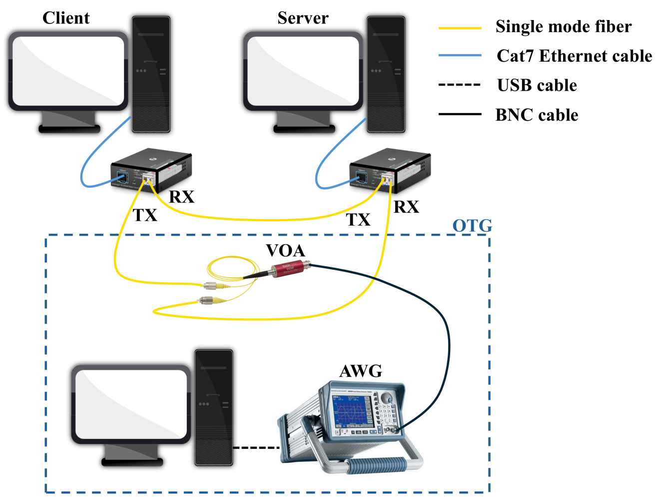



Fig. 1. Schematic of the developed testbed for the emulation of OFLs impaired by turbulence and pointing jitter.

Both client and server PCs are equipped with 10GbE network interface cards (NICs), which are connected via Cat7 Ethernet cables to two media converters. The 10GbE transceivers under test, plugged into the media converters, utilize On-Off Keying (OOK) modulation at a 1310 nm optical carrier wavelength and employ direct detection at the receiver. Although the setup supports a 10 Gbit/s maximum transmission speed, memory speed limitations forced us to reduce the traffic generation rate to 1 Gbit/s.

Fading is replicated on the forward link (from client to server) using a variable optical attenuator (VOA) -based optical turbulence generator (OTG). The SMF-coupled VOA dynamically attenuates the input beam intensity based on the applied voltage, which we externally generate by means of an arbitrary waveform generator (AWG) controlled via Matlab. By varying the VOA control voltage, we modulate the output signal's power, thereby emulating temporal fading patterns observed at OFL receivers. To this end, we implement algorithms capable of generating fading vectors that match the statistical and spectral characteristics of turbulence- and jitter-induced fading. Details about these algorithms are provided in the next subsection.

As described in [7], a commercial Micro-Electro-Mechanical Systems (MEMS)-based VOA allows us to reproduce attenuations ranging from 0.4 dB to 40 dB (0 V to 4.5 V control voltages) at the operating wavelength. A calibration procedure, detailed in [7], is employed to compensate for the VOA nonlinearities. The resulting dynamic range, combined with the DC-to-1 kHz VOA bandwidth, enables the developed OTG to replicate typical OFL fading conditions.

### A. Fading generation algorithms

Turbulence-induced scintillation and random pointing jitter cause irradiance fades at the receiving telescope, leading to power fluctuations at the receiver front-end. Received

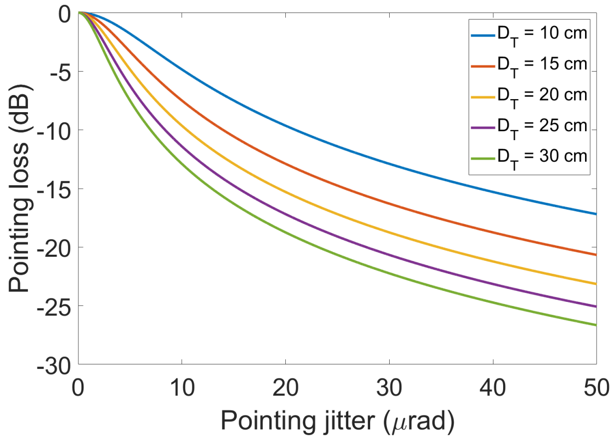



Fig. 2. Average pointing loss as a function of rms pointing jitter for different transmitter aperture diameters.

power time series can be efficiently synthesized based on the knowledge of the probability density functions (PDFs) and power spectral densities (PSDs) characterizing scintillation and jitter [11], [12]. Since scintillation and jitter are independent processes, their respective fading time series can be multiplied to generate a comprehensive received power time series that accounts for both impairments [12].

Fading time series are synthesized using the method developed by Gujar et al. [13]. This approach involves linear filtering of a zero-mean, unity-standard-deviation Gaussian random signal to produce an intermediate Gaussian signal that follows the desired PSD. Subsequently, a nonlinear transformation shapes the signal to match the target PDF.

For scintillation time series generation, we employ the lognormal PDF, which accurately models received power data from downlink field trials characterised by weak turbulence or significant aperture averaging [6], [7]. A low-pass model is assumed for the PSD, with cutoff frequency and roll-off slope analytically estimated as detailed in [11]. Further details can be found in our previous paper [7].

Jitter-induced fading time series generation is based on the algorithm developed by Giggenbach et al. [12], with minor modifications. Satellite microvibrations, occurring during nominal on-orbit operations, induce random pointing jitter, which is statistically described by Gaussian distributions for both the azimuth and elevation errors. If these errors are uncorrelated and identically distributed, the resulting radial pointing error follows a Rician distribution. In the absence of bias error, the Rician distribution reduces to a Rayleigh PDF [6], [14]. In this last case, the irradiance $I$ follows the beta distribution, with parameter $\beta = w_0^2/4\sigma_{jitt}^2$. Specifically, $w_0\ [rad]$ is the $1/e^2$ half-divergence angle and $\sigma_{jitt}\ [rad]$ is the root mean square (rms) angular pointing error.

The beta-distributed received power implies an average pointing loss given by $L_{pointing} = \beta/\beta + 1$. As illustrated in Fig. 2, this loss worsens as the pointing jitter increases and/or the transmitting telescope aperture widens.

Regarding the PSD of platform microvibrations, numerous space and ground measurements have been documented in the literature [14], [15]. Based on these measurement outcomes, the European Space Agency (ESA) adopted the following PSD as a specification for the Semiconductor Laser Intersatellite Link Experiment (SILEX)

$$S(f) = \frac{160}{1+f^2} \tag{1}$$

which corresponds to a rms value $\sigma_{jitt} \approx 16\ \mu rad$. A similar model was developed by the Japan Aerospace Exploration Agency (JAXA) for the PSD of transient caused by thruster firing [15]. Microvibration measurements for the LEO Optical Inter-orbit Communications Engineering Test Satellite (OICETS) are reported in [14]. The PSDs measured during the tracking and slewing modes for four downlink OFLs show good agreement with other space demonstrations and analytical models, including the one presented in (1).

Given the above considerations, jitter-induced fading time series are synthesized by generating two independent, zero-mean, unity-standard-deviation Gaussian random vectors, $X_1$ and $X_2$, representing orthogonal pointing error angles. These signals are low-pass filtered using first-order Butterworth filters with 1 Hz cutoff frequencies to shape their PSD according to (1). The filtered signals are then renormalized, obtaining $X_{1,n}$ and $X_{2,n}$, and converted into radial angular pointing error as $\theta_{jitt} = \sigma_{jitt}\sqrt{X_{1,n}^2 + X_{2,n}^2}$ . Finally, considering that the far field irradiance distribution $I(L,\theta)$ at distance L of a Gaussian beam can be expressed as

$$I(L,\theta) = I(L,0)\exp\left[-2\left(\frac{\theta}{w_0}\right)^2\right] \tag{2}$$

where $I(L,0)\ [W/m^2]$ is the irradiance at the beam center and $\theta\ [rad]$ is the angular deviation from the beam center, the beta-distributed time series of received power is computed as

$$P_{pointing}(t) = \exp\left[-2\left(\frac{\theta_{jitt}}{w_0}\right)^2\right] \tag{3}$$

## III. LEO DOWNLINK OFL FADING TIME SERIES GENERATION

In this section, we define the emulated OFL scenario and compute the corresponding fading time series. Specifically, we emulate a downlink transmission from a LEO satellite orbiting at an altitude of 600 km to an optical ground station (OGS) installed at an altitude of 10 m. The satellite pass was simulated in Matlab using a Simplified General Perturbations-4 (SGP4) propagator. By adjusting the satellite circular orbit and OGS coordinates, elevations up to 90° are achieved during the visibility period. The emulated scenario was limited to elevations above 20° due to practical link setup challenges at lower angles. Figure 3a shows the link distance and elevation angle as a function of time, while Fig. 3b depicts the slew rate originating from the satellite motion. The slew rate, derived from the satellite azimuth and elevation coordinates

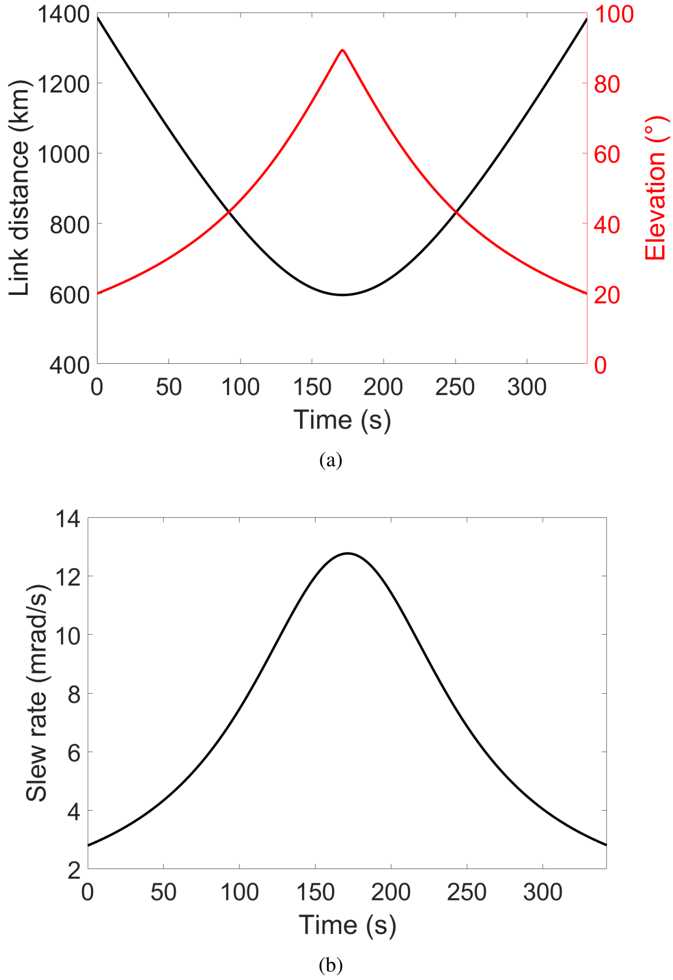



Fig. 3. (a) Link distance, elevation angle, and (b) slew rate as a function of time for the emulated LEO downlink OFL.

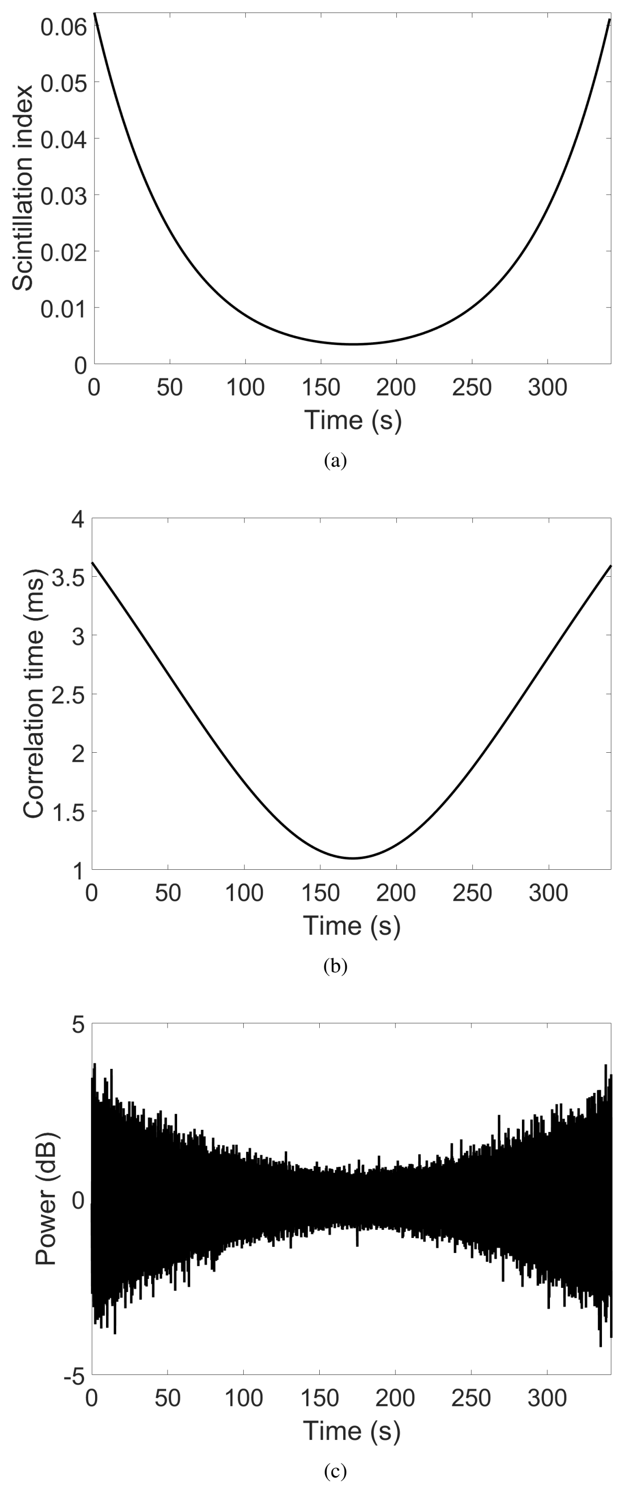



Fig. 4. (a) Scintillation index, (b) correlation time, and (c) fading time series as a function of time for the emulated LEO downlink OFL.

with respect to the OGS, introduces an apparent wind speed that impacts the scintillation coherence time.

Parameters for synthesizing turbulence-related fading time series at 1550 nm are presented in Fig. 4a and 4b. In detail, Fig. 4a shows the scintillation index, i.e., the normalized irradiance variance, over time. This parameter was computed using our OFL link budget tool [6]. To this end, turbulence was modeled with the modified Hufnagel-Valley model, setting a ground refractive index structure parameter of $1.17 \cdot 10^{-13}$ $\mathrm{m}^{-2/3}$ and a ground wind speed of 8 m/s, which correspond to strong turbulence. Despite this, a 40 cm receiving telescope reduces irradiance fluctuations through aperture averaging. Consequently, scintillation indices remain low throughout the satellite pass, with higher values occurring at lower elevations.

Figure 4b depicts the scintillation coherence time, calculated using analytical models. This parameter is crucial for determining the scintillation PSD cutoff frequency [11]. Shorter coherence times are observed at higher elevations. This correlates with the slew rate pattern, as faster slew rates induce increased apparent wind speeds, thus shortening scintillation coherence times.

The scintillation index and coherence time were used as inputs for the scintillation time series generation algorithm. To capture time-varying fading conditions, the LEO satellite pass was sampled at 1 s intervals. For each interval, we generated scintillation time series using the scintillation index and coherence time measured at the interval start point. Then, these series are concatenated to emulate the whole satellite pass. The resulting scintillation time series of normalized received power is shown in Fig. 4c. As expected, greater power fluctuations are observed at the beginning and end of the pass.

Pointing jitter levels, ranging from 1 $\mu rad$ to 3 $\mu rad$ in 0.25 $\mu rad$ increments, were emulated by generating nine distinct jitter-induced fading time series. These values reflect experimentally measured residual pointing jitters [16]. A transmitter aperture diameter of 15 cm was assumed onboard the satellite. The computed jitter-induced fading time series are shown in Fig. 5a, their histograms and corresponding theoretical beta curves in Fig. 5b, and their PSDs in Fig. 5c. As expected, deeper fades are observed at higher rms pointing errors. The time series mean values align with the theoretical average losses extrapolated from Fig. 2. Moreover, Fig. 5b confirms the match between the histograms and the beta curves, while Fig. 5c validates the desired low-pass behavior.

The overall time series of normalized received power, obtained by multiplying the scintillation- and jitter-related time series, is shown in Fig. 6.

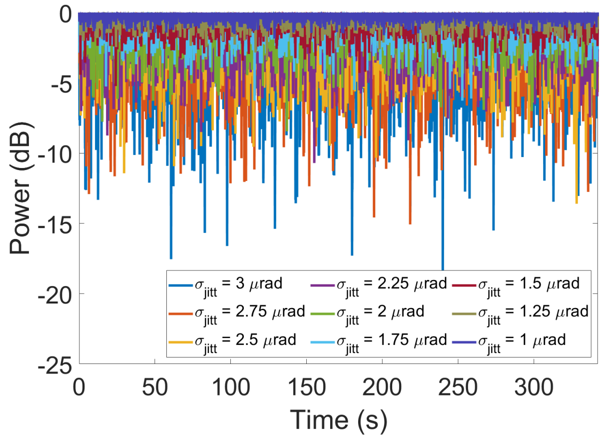
## IV. Experimental results

We conducted frame error rate measurements for each fading time series in Fig. 6, with each test lasting 342 s, i.e., the duration of the satellite pass. At a transmission rate of 1 Gbit/s, each test involved the transmission of about $2.9 \cdot 10^7$ Ethernet frames.

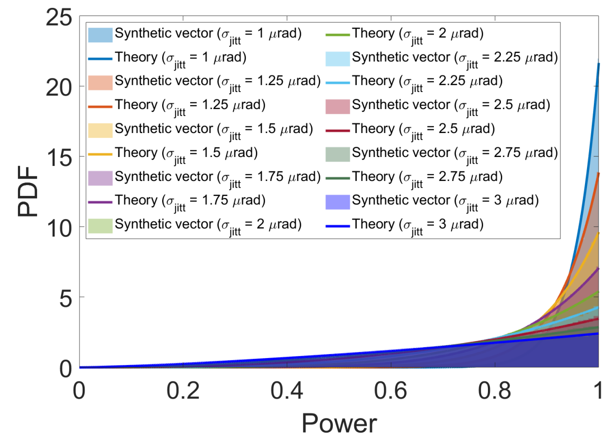
To perform these tests, the received power fading time series were converted into attenuation time series. For this purpose, we evaluated the transceivers' sensitivity, i.e., the received power level at which frame losses start to occur. By applying increasing constant attenuations via the VOA, we verified that losses appeared at an attenuation of 16 dB. Based on this outcome and considering a 1 dB margin, normalized powers were converted into attenuation values by associating a normalized power of 0 dB with a 15 dB attenuation. Subsequently, time series of VOA control voltages were computed based on the VOA transfer function, detailed in [7].

Figure 7 shows the frame error rate as a function of the rms pointing jitter. As expected, losses increase with jitter levels. This result stems from both the average attenuation and the fades caused by random pointing errors. Even in the absence of satellite microvibrations, turbulence-induced fading limits the achievable frame error rate to $2.3 \cdot 10^{-2}$ which, depending on specific application requirements, may not be an acceptable value. Indeed, lower frame error rates could be required. A steep increase in the frame error rate is observed with increasing jitter levels, reaching $6.6 \cdot 10^{-1}$ at $\sigma_{jitt} = 3\,\mu rad$. These results highlight the significant impact of pointing errors and emphasize the need for precise pointing and tracking systems.

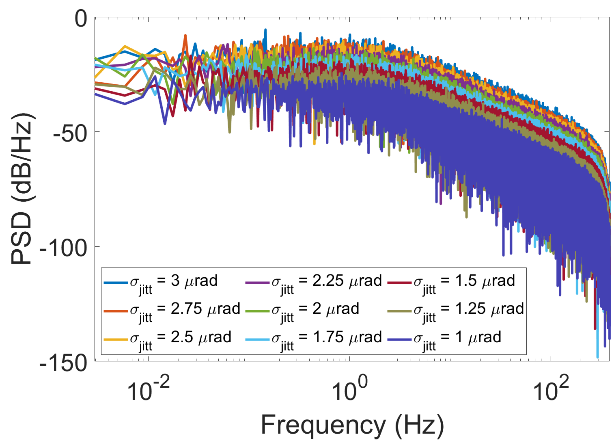
To enhance link availability, higher margins could be employed. However, technological limitations in transmitter power restrict the achievable margin. Thus, if reducing losses is the goal, the implementation of effective countermeasures should be considered.




(a)




(b)




(c)

Fig. 5. (a) Jitter-induced fading time series as a function of time for the emulated LEO downlink OFL. (b) Histograms of the jitter-related fading time series together with the corresponding theoretical beta PDFs. (c) PSDs of the jitter-related fading time series.

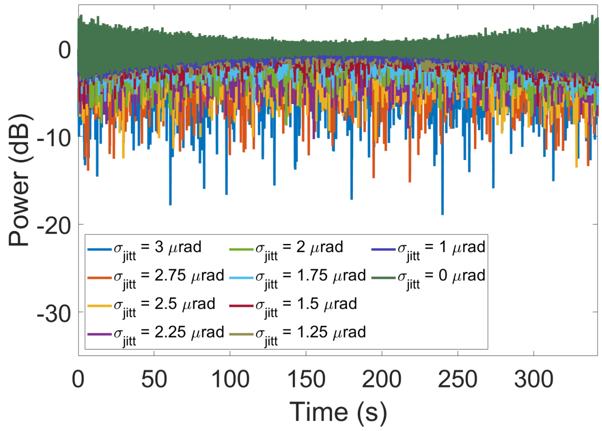



Fig. 6. Received power time series as a function of time for the scintillation- and jitter-affected emulated LEO downlink OFL.

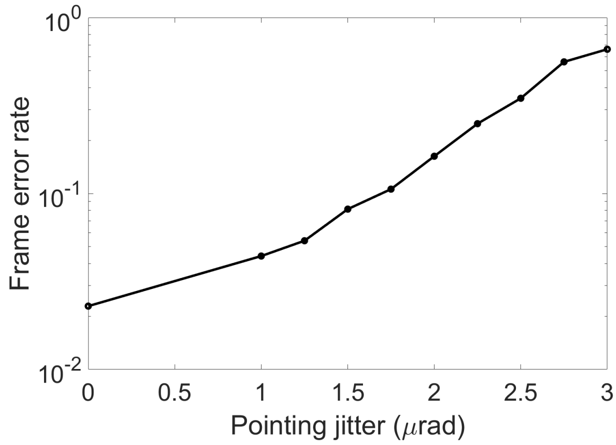



Fig. 7. Frame error rate as a function of the rms pointing jitter for the emulated LEO downlink OFL affected by scintillation and jitter.

## V. Conclusions

We presented an experimental testbed for emulating fading effects induced by turbulence and satellite microvibrations affecting OFLs. The setup relies on a compact and cost-effective VOA-based OTG. Computationally efficient algorithms were used to generate the fading time series. We demonstrated the algorithms' ability to accurately represent the statistical and temporal properties of both emulated phenomena, while accounting for the dynamics of LEO OFLs.

Finally, we experimentally evaluated the performance of a 10GbE downlink OFL using commercial fiber-based transceivers, under turbulence-induced scintillation and pointing jitter. The results highlight the stringent requirements that pointing and tracking systems must meet to achieve the desired link availability. Even without jitter, a degradation in link performances was observed due to turbulence.

In a future effort, the developed testbed could be employed to assist the design of mitigation techniques and countermeasures to ensure higher link availability.

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